

Experimental and numerical local heat transfer study on micro pin fin with tip clearance

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HIGHLIGHTS

- Local heat transfer downstream a micro pin fin with tip clearance was studied.
- Novel microfabrication techniques for manufacturing tip clearance and SU8-Pyrex bonding were developed.
- A 3-D conjugate CFD model were developed and revealed various fluid flow and heat transfer contours.
- Significant heat transfer enhancement for a sufficiently large tip clearance in the wake region was achieved.

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ABSTRACT

Fluid flow and local heat transfer in a microchannel with a 150- μm diameter pin fin with a tip clearance were experimentally and numerically studied for three Reynolds numbers in laminar regime. Tip clearances of 0, 30, 45 and 100 μm in a 200- μm high microchannel. Experimental and numerical local temperatures and the corresponding Nusselt numbers along the centerline of the pin fin were presented and discussed. Local temperatures were measured on top of the heater surface and downstream the pin fin through micro resistance temperature detectors (RTDs). A conjugate CFD modeling capable of simulating solid/fluid conduction and convection revealed velocity, heat flux and heat transfer coefficient over the heated surface. Nusselt number and wake length for a range of tip clearances were presented and compared with full-height pin fin. Experimental and numerical results showed that a tip clearance can significantly enhance heat transfer in the wake region. Simulations revealed that tip clearance alters the flow structure by increasing the three dimensionality of the flow, promoting mixing, shortening the wake region, and increasing the velocity downstream the pin fin. A tip clearance with a height of 100 μm was found to provide the best heat transfer enhancement.

1. Introduction

Thermal management of electronic chips in computers, laptops, cell phones and power inverter module in electric vehicles is a critical issue. The need for miniaturization coupled with the growing increase in power density of next generation electronic devices calls for more efficient cooling strategies. One notable approach is to extend the effective heat transfer surface area and promote fluid mixing by incorporating of pin fins into the cooling domain [1–13]. However, elevated pressure drop and high pumping power is a hindrance that needs to be carefully addressed. At the conventional scale, a clearance between the pin fin's tip and the channel wall (a.k.a tip clearance (TC)), depicted in Fig. 1e and f, has shown to alleviate the issue while maintaining a noticeable high convective heat transfer coefficient [14–16].

Sparrow et al. [15,16] were the first to systemically examine cylindrical pin fin arrays with various TC s, and observed a slight decrease

in the heat transfer coefficient in comparison to full pin fin. Mahajan [17] examined different TC gaps and suggested using TC at a minimum to maintain the thermal performance of the heat sink. Jubran and Hamdan [18] performed an experimental study with inline and staggered array of cylinders at Reynolds number ranging from $Re = 5 \times 10^3$ to 5.4×10^4 with TC of 33% and 50% of channel height and reported a 7% reduction in heat transfer in staggered configuration when the heat flux was applied on the bottom wall. Wirtz et al. [19] studied the effects of bypass flow in the TC region with air and also observed a reduction in heat transfer with TC clearance. Furthermore, they developed a correlation for this effect based on their experimental assessments. Reduction in thermal performance is less significant in cases with relatively smaller clearances. Moores and Joshi [20] ran a series of experiments with water at Reynolds number from 200 to 10,000 and considered a TC of 0–25%. They examined arrays with three different heights to diameter ratios between 0.5 and 1.1 with only the

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Nomenclature

A_c	microchannel cross-sectional area (m^2)
A_h	heater area (m^2)
D	pin fin diameter (m)
D_h	hydraulic diameter of microchannel (m)
h	heat transfer coefficient ($W/m^2.K$)
H	height of microchannel (m)
K_w	water thermal conductivity ($W/m.K$)
L_w	wake length (m)
L_w/D	dimensionless wake length ($-$)
$\dot{m}$	mass flow rate (kg/s)
Nu	Nusselt number ($-$)
P	wetted perimeter of microchannel cross-section (m)
$\dot{Q}$	electrical power (W)
$\dot{Q}_{loss}$	heat loss (W)

q''	net heat flux (W/m^2)
δR	uncertainty of R
Re	Reynolds number based on hydraulic diameter ($-$)
T_{RTD}	temperature measured by RTDs ($^{\circ}C$)
T_{ref}	fluid temperature ($^{\circ}C$)
TC	tip clearance (m)
TC/H	dimensionless tip clearance ($-$)
U	average velocity in microchannel (m/s)
μ	dynamic viscosity of water ($kg/m.s$)
ρ	density of water (kg/m^3)
ΔT	temperature difference ($^{\circ}C$)
X	parameters in uncertainty calculation
δX	equipment uncertainty
x/D	dimensionless longitudinal distance ($-$)
W	width of microchannel (m)

base heated and found that incorporating a TC smaller than 10% of increases the mean heat transfer coefficient by 15–23%, respectively. Also, they noted that 3-D effect induced by vortex generation offsets the lower heat transfer efficiency of the tip surface compared to a full pillar. More recently, Sastre et al. [21] conducted an experimental and numerical study to examine flow topology in finned heat sinks with TC s. They found that the Nusselt and Graetz numbers collapsed into multiple groups of curves rather than a single curve, and those families governed by specific geometry of the TC .

A few studies examined TC at the micro scale. Reyes et al. [22] used a variable TC in square-shaped microchannels and compared heat transfer and pressure drop with a full height pin fin. They reported a slight deterioration in heat transfer and a marked reduction in pumping power with TC . Mei et al. [23] numerically examined the effect of TC on heat transfer for a micro scale heat sink with inline cylindrical pin fin arrays. Various TC s from 0 to 1000 μm at $Re = 33$ –350 were modeled and it was found that the thermal performance of the heat sink is sensitive to TC at low Reynolds numbers. A non-circular fin arrays were studied by Rajabifar et al. [24] who developed a 3-D numerical model to investigate the thermal and hydrodynamic effect of TC s on top of a micro heat sink with array of oblique fins using Nano-Encapsulated Phase Change Material (NEPCM) particles in water. Tip clearance to channel width ratio of 0–0.74 for low Reynolds numbers below 230 were considered. For a TC to channel width ratio between 0.16 and 0.26, higher Nusselt numbers compared to array of full height fins were observed. Tabkhi et al. [25] numerically studied a single pin fin in a microchannel and compared heat transfer for various TC s values with a full pin fin. They varied the heater location and found that when the heater is on the opposite side of the TC , the heat transfer coefficient was higher. They also found optimum TC corresponding to a maximum heat transfer coefficient.

A TC between the pin fins and the wall introduces additional heat transfer and fluid flow processes. A fraction of the fluid travels above the tip, and as a result, the flow velocity through the pin fins decreases, which tends to hinder heat transfer from the pin fin surface area. Conversely, the tip provides additional surface area available for cooling. Furthermore, TC modifies the hydrodynamic of the flow in a way that can promote mixing and three dimensionality of the flow field, thereby enhancing heat transfer. An optimum TC can overcome the adverse effect of the bypass flow while at the same time leverage the additional heat transfer area and the increased flow mixing.

The reduction in the heat transfer of devices with large TC are mostly due to the decreased surface area available for heat transfer (i.e., shorter fins correspond to reduced heat transfer surface area). This might not be true at the micro scale. As the heat transfer coefficient increases with diminishing length scales, fin efficiency tends to diminish compared to their large-scale counterparts. It is therefore, reasonable to speculate that TC micro pin fin heat sinks has more potential than at the conventional scale. Yet, few studies have addressed TC at diminishing length scales. Furthermore, studies aimed to decompose the various effects caused by the TC are lacking.

This study investigates the effect of TC on the flow and heat transfer downstream a micro pin fin. Here, single pin fin was selected in order to decouple the interaction between adjacent pin fins. Conjugate conduction/convection effects were minimized by incorporating materials with low thermal conductivity.

2. Methodology

2.1. Micro device and microfabrication

The micro-device (Fig. 1) consisted of two SU8 negative resist layers sandwiched between two Pyrex substrates (more detailed information on SU8 is provided in [26]). The SU8 formed the microchannel, the pin fins, and the TC —one layer was used to fabricate the pin fins while two layers, including the pin fin layer, formed the channel's sidewalls. On the bottom Pyrex wafer, photolithography, metallization, and etching processes were carried out to make the heater and resistance temperature detectors (RTDs). The top pristine Pyrex substrate was bonded on top of the SU8 layer through a clear 125- μm thick silicone adhesive tape.

A 150- μm diameter single circular pin fin was placed in a micro-channel with dimensions of $L \times W \times H = 23 \text{ mm} \times 1.5 \text{ mm} \times 0.2 \text{ mm}$. Multiple devices with different pin fin heights were fabricated to produce a range of TC s on top of the pin fin. (The pin fin was placed 13 mm downstream the inlet.) The heater's leading edge was positioned 1.425 mm upstream the pin fin (Fig. 2a) and covered the entire width of the channel except for a region around the pin fin. A set of seven, 55 $\mu m \times 55 \mu m$, RTDs were positioned along the microchannel surface on top of the heater (Fig. 2). The rectangular heater and serpentine RTDs were made from a 30-nm thick Platinum layer and were electrically connected to contact pads through 1- μm thick Aluminum vias (Fig. 2b and

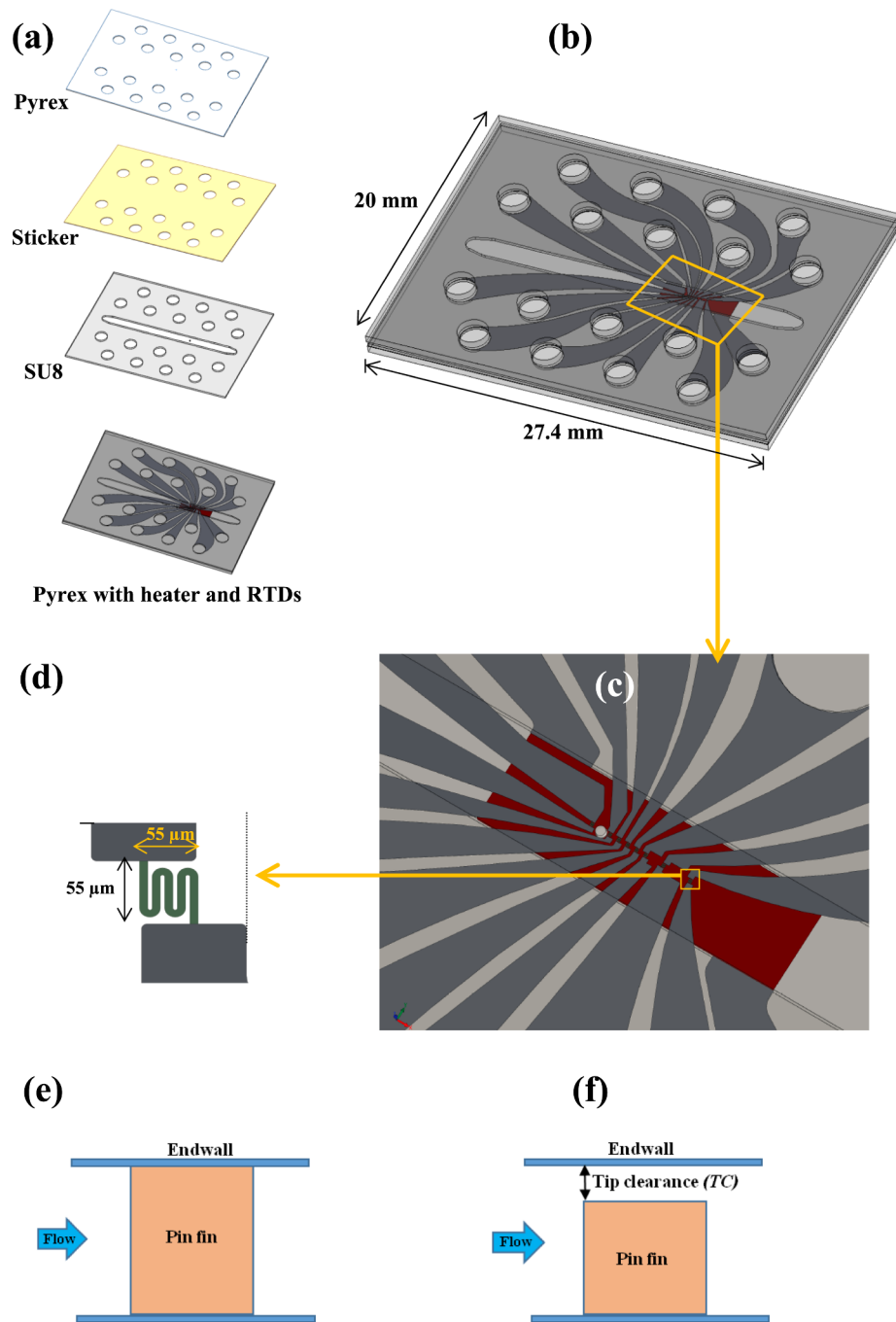


Fig. 1. CAD drawing of the micro-devices. (a) A blowup schematic showing the four primary layers of the device including a bare Pyrex substrate to seal the microchannel from the top, a microchannel and a pin fin made from SU8, an adhesive layer, and another Pyrex substrate with the heater, RTDs, and vias; (b) the four layers bonded together; (c) zoom-in view of the heater, pin fin, RTDs, and vias; (d) Zoom-in view of the serpentine RTD. (e) Schematic side-view of a full pin fin and (f) Schematic side-view of a pin fin with TC.

c).

To fabricate the micro device, a 500- μm thick Pyrex wafer was initially piranha cleaned using Hamatech tool (Fig. 3a). A 0.5- μm layer of Si_3N_4 was then deposited on the top surface using Oxford PECVD (Fig. 3b), and subsequently a 7-nm Titanium and then a 30-nm Platinum were deposited using an AJA Sputter (Fig. 3c). Titanium was deposited to enhance the adhesion of the Platinum on the Pyrex substrate. Photolithography process followed by AJA Ion mill were used to pattern the platinum and form the heater (Fig. 3d). To fabricate the electrical vias of the heater, Titanium and Aluminum were deposited on top of the Platinum (Fig. 3e). An additional photolithography and wet

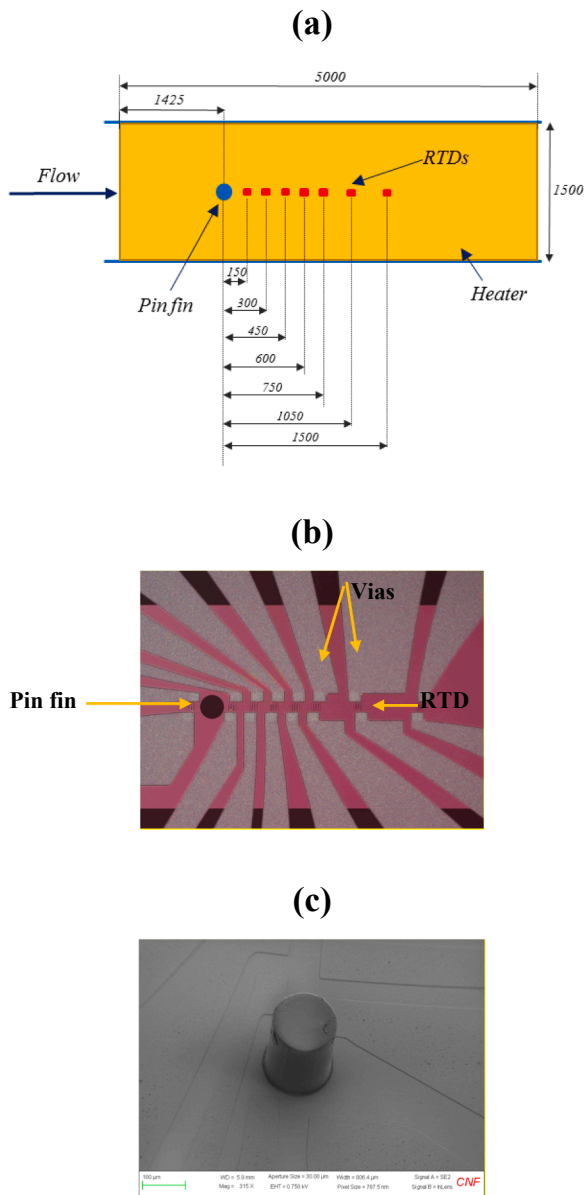


Fig. 2. (a) Schematic drawing of the heater—the yellow rectangle with the RTDs (shown as small red square). Dimensions are in μm . RTDs dimensionless distances from the center of pin fin are. $x/D = 1, 2, 3, 4, 5, 7$ and 10 . (b) Microscopic image of the device showing the pin fin, heater, RTDs and vias. (c) SEM image of the pin fin with a TC. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

chemical etching processes were carried out to pattern the Titanium/Aluminum layers and to form the electric vias (Fig. 3e–g). To insulate the heater and its aluminum vias from the RTDs, a 1- μm layer of SiO_2 was deposited on top of the aluminum using OXFORD PECVD tool (Fig. 3h). Subsequently, the RTDs and their vias were fabricated in a sequence of repetitive processes similar to those used for the heater and its vias (Fig. 3i–m). To insulate the two metal layers from the fluid, an additional 500-nm SiO_2 layer was deposited using OXFORD PECVD on top of the RTDs and their connections (Fig. 3n). In order to provide electric connections between the heater and RTDs' vias, contact pads were formed by etching the SiO_2 in two separate steps with different depths during a dry etch process using OXFORD 82 (Fig. 3o and p).

To fabricate a range of *TCs* between the pin fin and the upper wall, two separate layers of SU8 with two different masks were used. The layer adjacent to the processed Pyrex substrate included the channel sidewalls and the pin fins, while the second layer only included the channel sidewalls. Depending on the height of each layer, different series of SU8 were used. SU8 100 was used for the first layer and a thinner layer of SU8 (SU8 50 and SU8 2000) was used for the second layer. The fabrication of the SU8 included spinning, soft bake, exposure (patterning using photolithography), post-exposure bake and development. Typically, hard-bake was used, but to fabricate two layer of SU8 with different patterns on top of each other, modifications were required. For the first layer, the SU8 100 was placed and spun on the bottom wafer. After a soft bake process of a couple of hours, the SU8 was masked and exposed to an ultraviolet (UV) light to pattern the first SU8 layer. In preparation for the second SU8 layer, the post-exposure bake of the first layer was avoided and another SU8 layer was processed on top of the first one. The second layer was soft baked on a hot plate simultaneously with the post-exposure bake of the first layer. The second layer was then exposed to UV light through its designed photolithography mask and placed again on a hot plate for a post-exposure bake. Developing was done at the same time for both SU8 layers after completing the hard bake of the second layer. By placing the wafer in an SU8 developer, rinsing, and drying the wafer, the microchannel and the pin fins with *TCs* were formed. A profilometer was used to measure the height of the *TC* on top of the pin fins. In order to bond the SU8 to the top Pyrex layer, the bottom Pyrex and the SU8 were diced into separate devices. Additionally the pristine Pyrex wafer and a double-side, 135- μm silicone adhesive sticker were also diced to the same dimension of the individual devices. Subsequently, the sticker was placed on the SU8 side of the device to protect the pin fins during the drilling process. Inlet and outlet holes were drilled through the bottom Pyrex substrate using traditional machining processes. Finally, the top glass layer was placed on top of the sticker. By varying the thickness of the first and second layers of the SU8 while keeping a same height of the final device (channel height), pin fins with different *TCs* were fabricated. Fabricating a constant channel height was necessary for an adequate comparison.

2.2. Experimental apparatus and procedure

2.2.1. Experimental loop

The open fluidic loop setup is shown in Fig. 4. An upstream water reservoir was connected to a high-pressure Nitrogen tank through a pressure regulator. This allowed maintaining the reservoir at a desired pressure and supporting a pressure difference to propel the flow through the micro device. Before entering the package and micro device, the water passed a filter and a rotameter (FL-3805ST, OMEGA®). Pressure transducers (PX-209-30V45G5V, OMEGA®) and T-type thermocouples were installed upstream and downstream the package and were connected to a compact data acquisition system (NI-cDAO 9178)

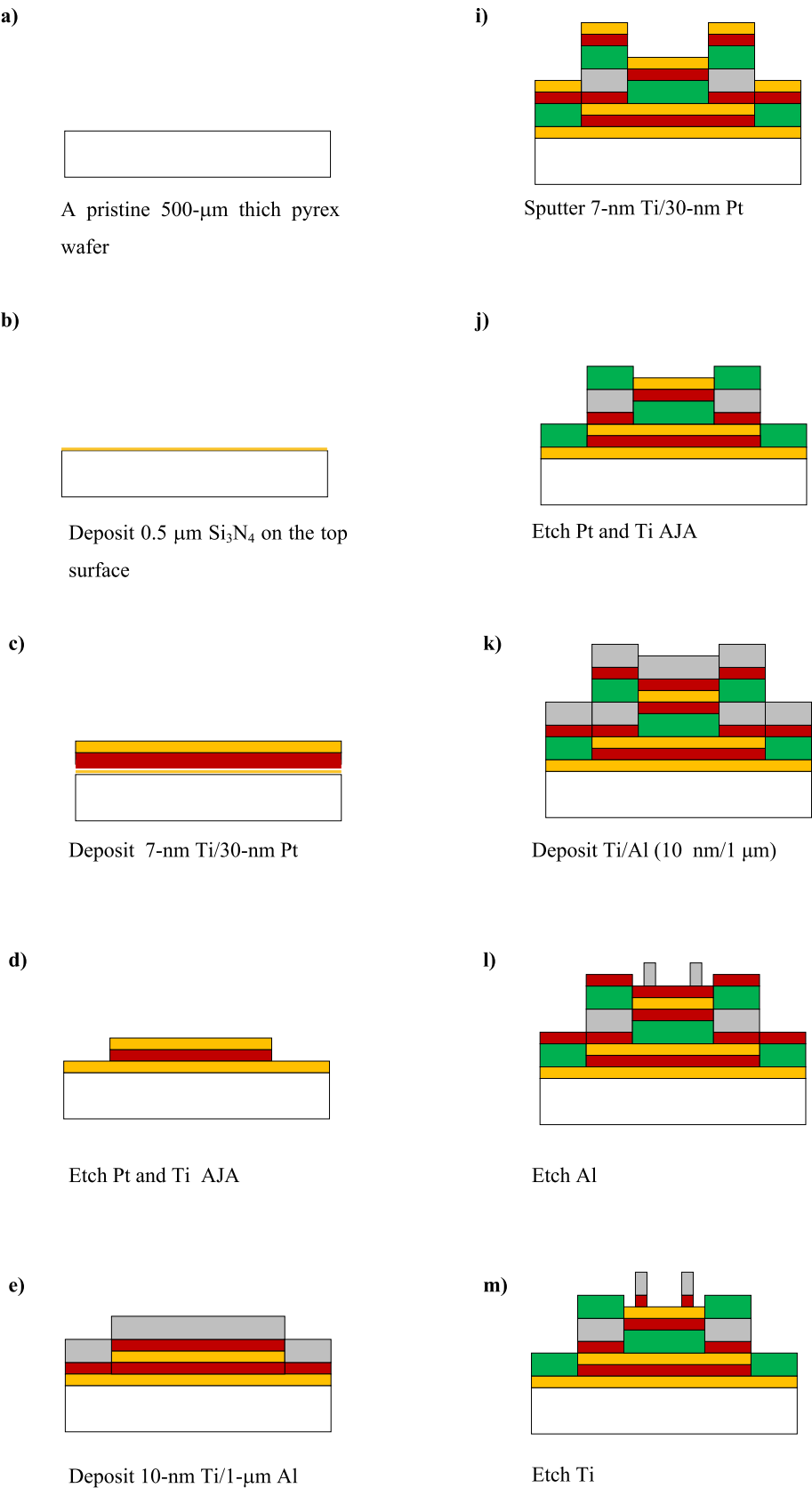


Fig. 3. Microfabrication processes on the bottom Pyrex substrate.

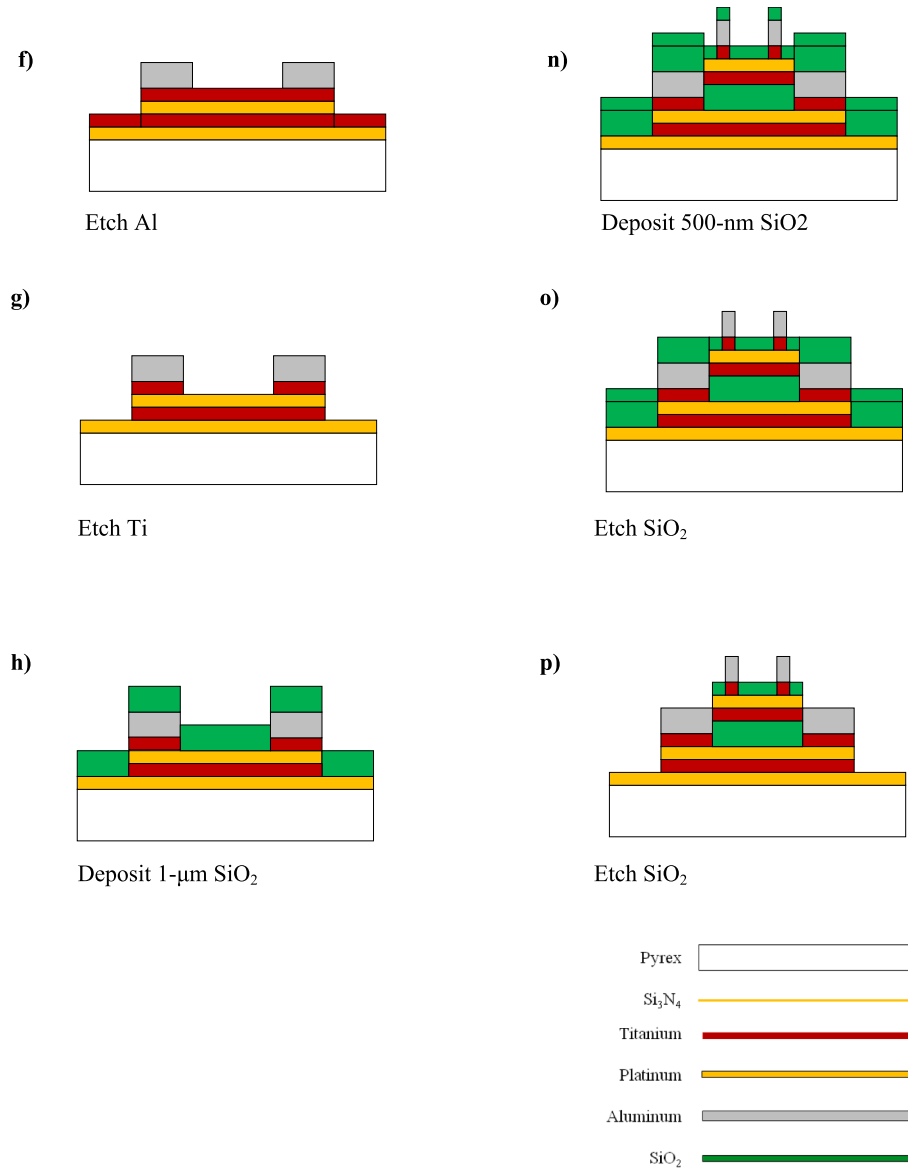


Fig. 3. (continued)

and a LabVIEW program. A high-speed camera (Phantom Miro 320, Vision research®) together with an inverted microscope (Axio Observer®, ZEISS) were used to monitor the device condition during experiments.

2.2.2. Package, printed circuit board, and electrical connections

The device was placed in a special octagonal package, 100 mm long and 22 mm high, made of Polyoxymethylene (a.k.a., Delrin) that enabled connection to the experimental flow loop, data acquisition unit, and power supply (Fig. 5). A pocket was made into the package to house the micro device. On the bottom of the pocket, holes were made that served as an inlet and an outlet for the flow. O-rings were embedded into these holes for sealing and support purposes. In order to secure the device in the package, a cover plate was also machined from

Delrin and bolted on top of the device. A custom-made printed circuit board (PCB) was installed on top of the cover plate to provide electrical connections between the device and the electrical equipment through spring loaded probe pins, which connected the device's heater and RTDs to a breakout board through ribbon cables. Breakout board connected two multimeters (Agilent 34410A®), a power supply (Keysight, E3645A), and the data acquisition system to the micro device. The resistances of the RTDs were measured through a signal conditioning extension for instrumentation (SCXI) data acquisition unit from National Instrument (NI®).

2.2.3. Data reduction and measurement uncertainties

Reynolds number based on the hydraulic diameter of the micro-channel was calculated according to:

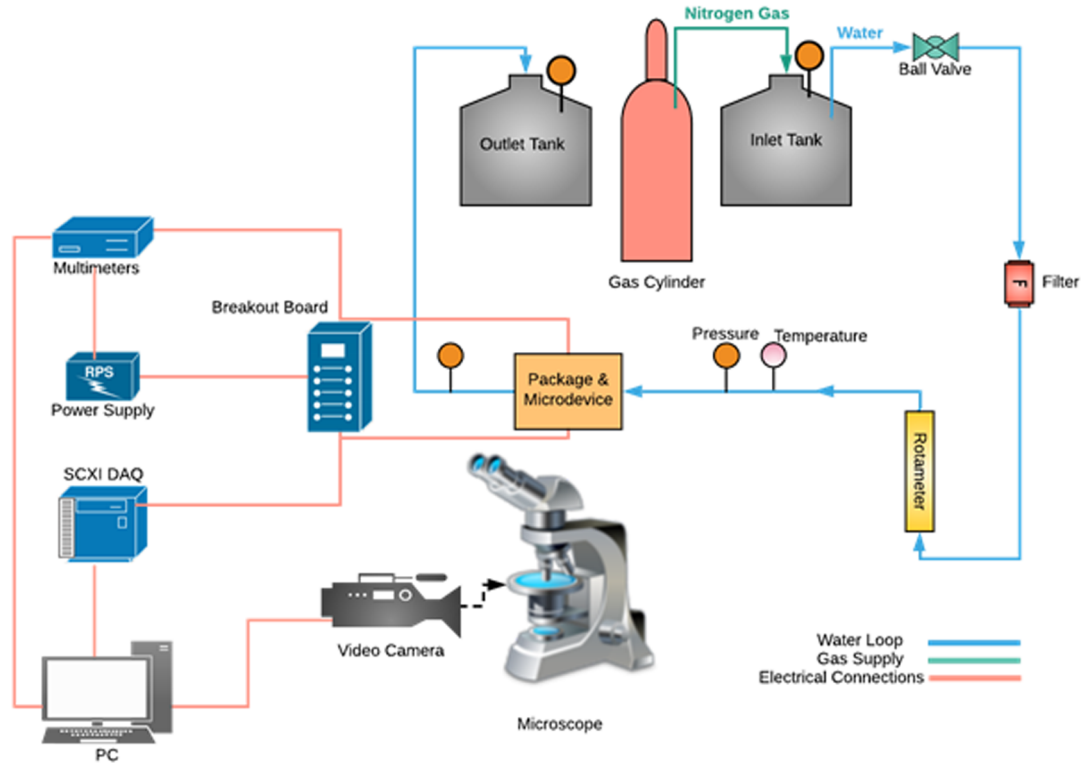


Fig. 4. Schematic of experimental apparatus.

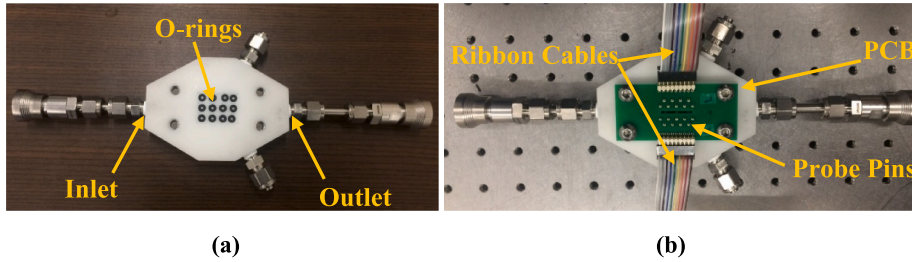


Fig. 5. (a) The package that provided fluid and electrical connections between the micro device and the experimental apparatus. The micro device was placed on top of a set of O-rings. (b) Printed circuit board connected the probe pins to the contact pads on the bottom Pyrex wafer, to the cover plate, and to the ribbon cables. Swagelok connectors were used to connect the package to the test loop.

$$Re = \frac{\rho U D_h}{\mu} \quad (1)$$

where U is the mean velocity in the microchannel defined as:

$$U = \frac{\dot{m}}{\rho A_c} \quad (2)$$

$\dot{m}$ is the mass flow rate and A_c is the cross-sectional area of the microchannel calculated by $W \times H$.

The hydraulic diameter of the channel ($D_h = 353 \mu\text{m}$) was defined by $\frac{2WH}{W+H}$, where W and H were the width and height of the microchannel

Table 1
Test matrix explaining TC values and Reynolds numbers.

Device	Reynolds number		
	395	627	827
TC = 0 (full pin fin)	EXP1	EXP5	EXP9
TC = 30 μm	EXP2	EXP6	EXP10
TC = 45 μm	EXP3	EXP7	EXP11
TC = 100 μm	EXP4	EXP8	EXP12

rectangular cross section.

Heat flux from the heater, q'' , was calculated by dividing the total heating power supplied to the heater ($\dot{Q}$) minus heat loss ($\dot{Q}_{\text{loss}}$) by the heater area (A_h):

$$q'' = \frac{\dot{Q} - \dot{Q}_{\text{loss}}}{A_h} \quad (3)$$

The heater's power was calculated by multiplying the voltage (V) and the current (I) through the heater. The heat flux was set to $q'' = 53 \text{ W/cm}^2$.

Local heat transfer coefficient at the various RTD locations were calculated according to:

$$h = \frac{q''}{\Delta T} \quad (4)$$

where ΔT is the temperature difference between the RTD and the fluid temperatures ($T_{\text{RTD}} - T_f$).

Accordingly, local Nusselt number was defined by Eq. (5):

$$Nu = \frac{h D_h}{k_w} \quad (5)$$

where k_w is water thermal conductivity.

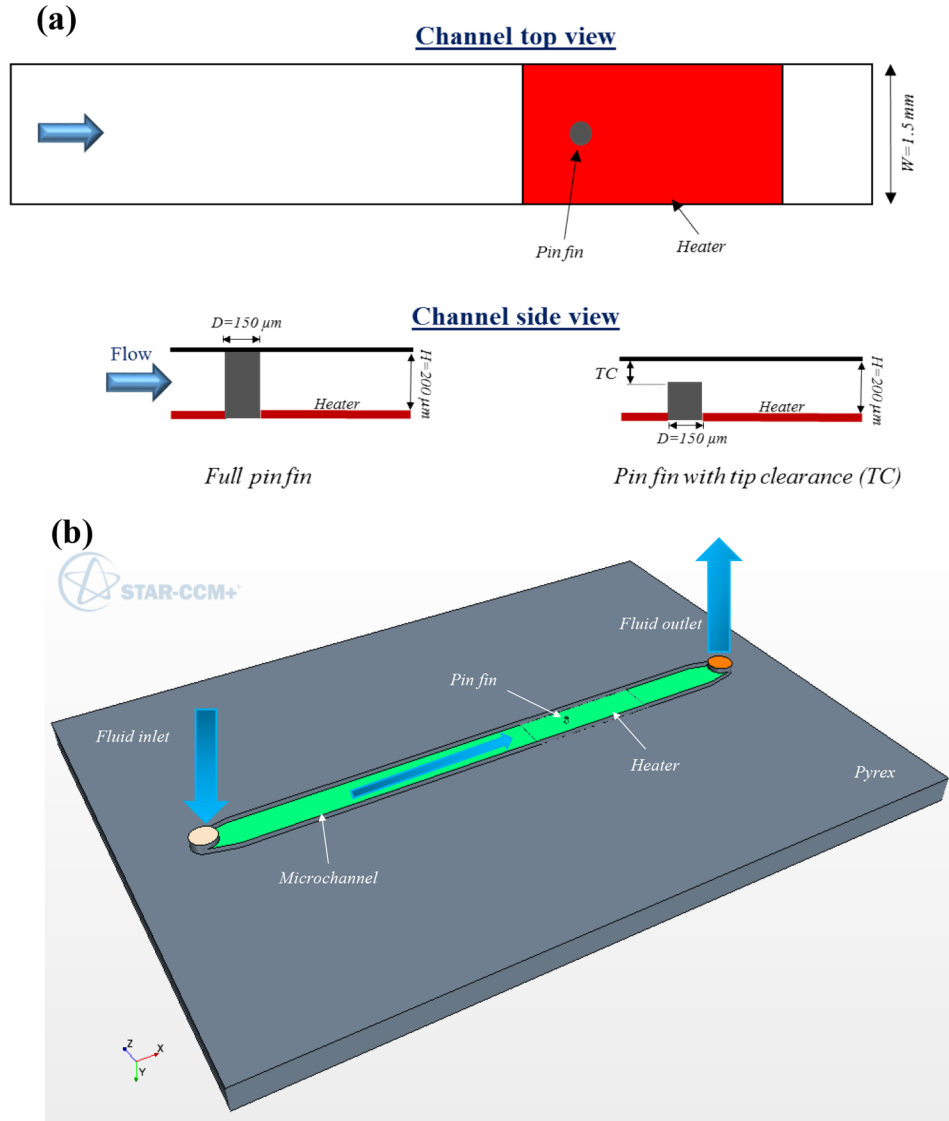


Fig. 6. (a) Schematics top and side-views of the simulated configurations (b) Schematic of the numerical domain denoting fluid inlet, outlet, pin fin location, heater and Pyrex.

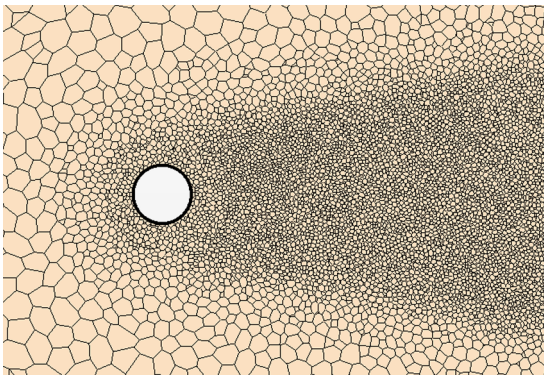


Fig. 7. Computational grid around the pin fin.

Measurement uncertainties were calculated based on the standard method discussed by Moffat [27]. The uncertainty δR depends on the variables $X_1, X_2, \dots, X_n$ is defined by Eq. (6):

$$\delta R = \sqrt{\sum_{i=1}^n \left(\frac{\partial R}{\partial x_i} \delta x_i \right)^2} \quad (6)$$

In the above equation, the partial derivative of R with respect to X_i , $\frac{\partial R}{\partial x_i}$, is the sensitivity coefficient for the result R with respect to the measurement X_i . δx_i is the parameter X_i uncertainty, which was provided by the manufacturers' datasheets.

Following the above equation, the measurement uncertainties for each data point were calculated. Resistances of the RTDs were measured using a DAQ (NI-SCXI 1000, NI®) with an uncertainty of $\pm 0.5 \Omega$. Consequently, uncertainty of the measured temperatures was $\pm 1^\circ \text{C}$. The uncertainty in the heat transfer coefficients and Nusselt numbers

were less than 6%. Reynolds number uncertainty was less than 5%.

2.2.4. Experimental procedure

A custom-made copper plate and an insulated oven were used to calibrate the RTDs' temperatures. To obtain the temperature-electric resistance relationship for each RTD (a.k.a., the calibration curve), one module (NI-SCXI 1300) in the NI-SCXI 1000 DAQ was used to generate a current of 100 μ A while the second module (NI-SCXI 1303) measured voltage for each RTD. After reaching steady state, the temperature of the RTD and its resistance were recorded. The process was repeated for different temperatures and a calibration curve for each RTD was obtained. Results showed a linear dependency between resistance and temperature.

The heater's power was increased in steps and after reaching steady state, the RTDs' signals downstream the pin fin were recorded for about 30 s with a sampling rate of 1000 Hz. Reynolds numbers of 395, 627, and 827 based on the hydraulic diameter of the channel (353 μ m) were studied with a heat flux of $q'' = 53$ W/cm². The effect of TCon the heat transfer coefficient was studied in four devices with a single circular pin

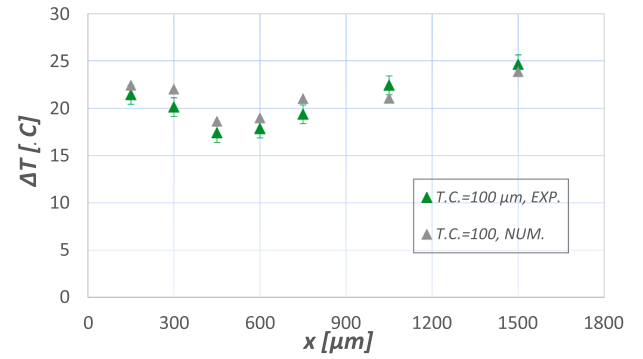


Fig. 9. Comparison between numerical simulation and experimental measurement.

fin for TCs of 0, 30, 45 and 100 μ m. Table 1 lists the 12 different experiments (EXP1 through EXP12).

2.3. Numerical model

The numerical simulation was performed using a commercially available software, STAR-CCM+®. Fig. 6a schematically depicts the simulated configurations with and without TC. The computational domain consisted of the microchannel, the pin fin, the heater, and the Pyrex substrate (Fig. 6b). The bottom surface of the microchannel was in direct contact with the heater and the Pyrex. To simulate the heat transfer to the fluid in the microchannel, two interfaces were created, which included the heater/microchannel and the Pyrex excluding the heater/microchannel. Although conduction heat transfer from the heater to its adjacent Pyrex substrate was numerically estimated to be less than 0.1%, the model was capable to account for it and estimated the heat transferred from the Pyrex to the fluid outside the heater area.

2.3.1. Fluid section

Previous studies with the same pin fin and microchannel dimensions have shown that the flow is laminar when $Re \leq 950$ (Re is based on channel hydraulic diameter) [28]. Since the tested Reynolds numbers were lower than 950, all simulations were conducted using laminar model. Inlet velocities and temperatures obtained from the experiments were considered for the microchannel inlet. The velocity was calculated based on the chosen Reynolds number at a designated inlet temperature. The microchannel outlet pressure was set to atmosphere, and the outlet temperature was calculated by the model during the simulation. The pin fin and the microchannel walls were assumed to be adiabatic because of the low thermal conductivities of SU-8 and Pyrex. The segregated flow model was utilized to solve four equations including three equations for the velocity in the x, y and z directions and one equation for the pressure through an uncoupled process. Continuity and Navier-Stokes equations were linked via a predictor-corrector approach. SIMPLE-type algorithm was used in conjunction with Rhie-and-Chow-type pressure-velocity coupling. A second-order upwind convection scheme was chosen for the convection terms. Boundary secondary gradients for diffusion and flow-boundary diffusion fluxes were activated for each simulation. The total energy equation was solved by a segregated fluid temperature model to obtain temperature as unknown variables. The equation of state was used to calculate the enthalpy from the previously known temperature. Similar to the flow equations, a second-order scheme for the convection terms together with boundary

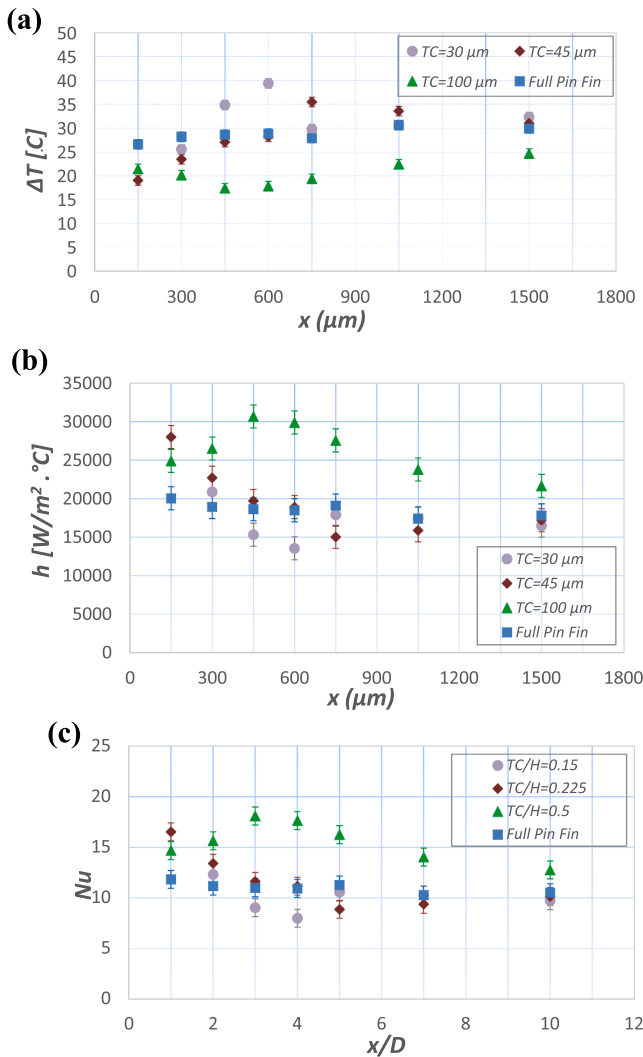


Fig. 8. (a) Temperature differences at $q'' = 53$ W/cm² (b) Local heat transfer coefficient and (c) Local Nusselt number, all for $Re = 395$.

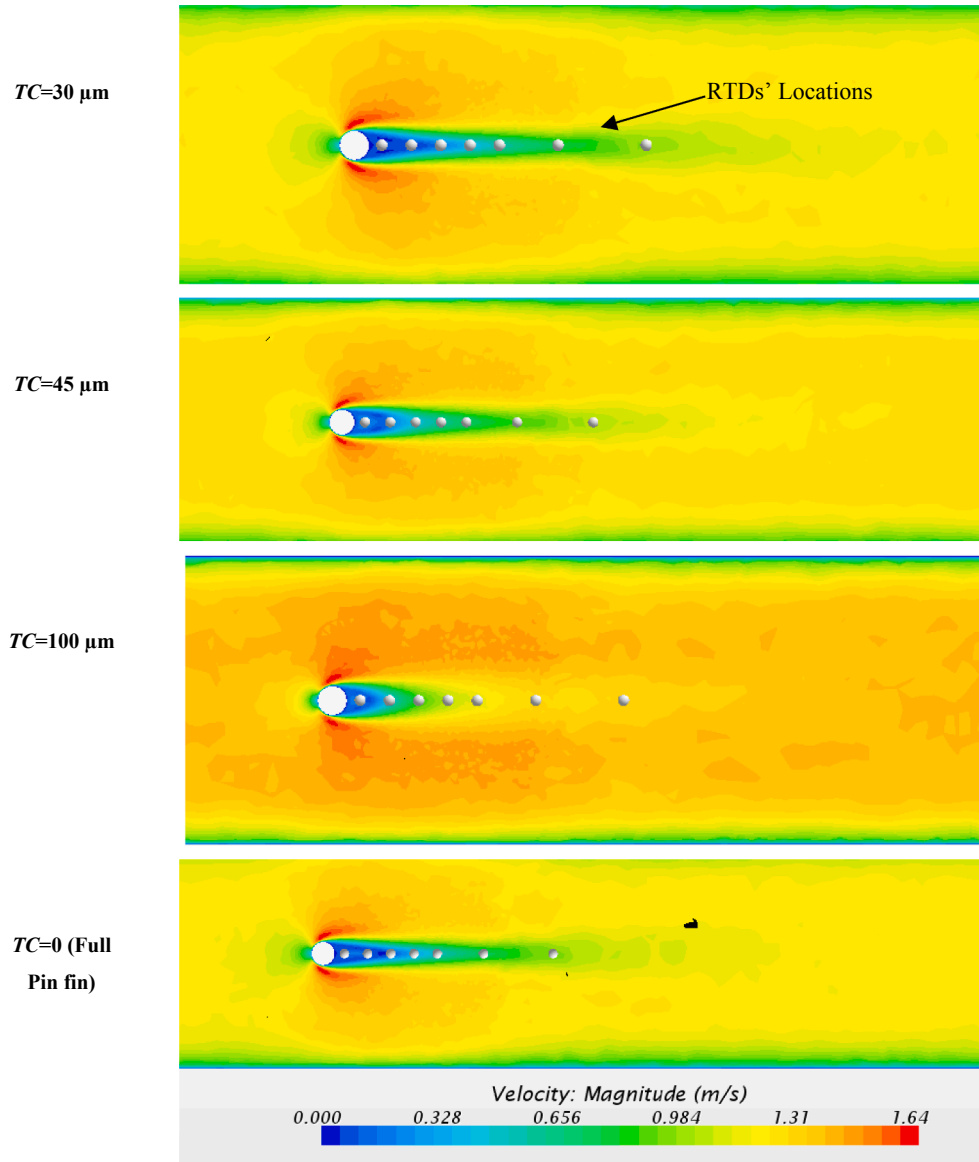


Fig. 10. Contour of velocity magnitude 50 μm away from the heater for $Re = 395$.

secondary gradients for diffusion and flow-boundary diffusion fluxes were employed to solve the energy equation.

2.3.2. Solid section

For the Pyrex, segregated solid energy and steady state model were activated. Since one side of the Pyrex was exposed to the environment, external natural convection heat transfer coefficient of $h = 5\text{ W/m}^2 - \text{K}$ was assigned.

Heating element made of Platinum was embedded inside the Pyrex. Therefore, the heater was simulated as a very thin shell with a thickness

of 30 nm placed inside the Pyrex. Homogenous heat generation source was defined inside the heater. Segregated solid energy and steady-state model were solved in the heater. The defined interfaces inside the microchannel, Pyrex, and heater were enabled for transferring heat from the heater to the Pyrex and the fluid.

2.3.3. Grid generation

Hybrid grid with prism layers on the pin fin and polyhedral elements were used (Fig. 7). Three sets of computational domains with coarse, medium, and fine grids were generated by varying the grid base

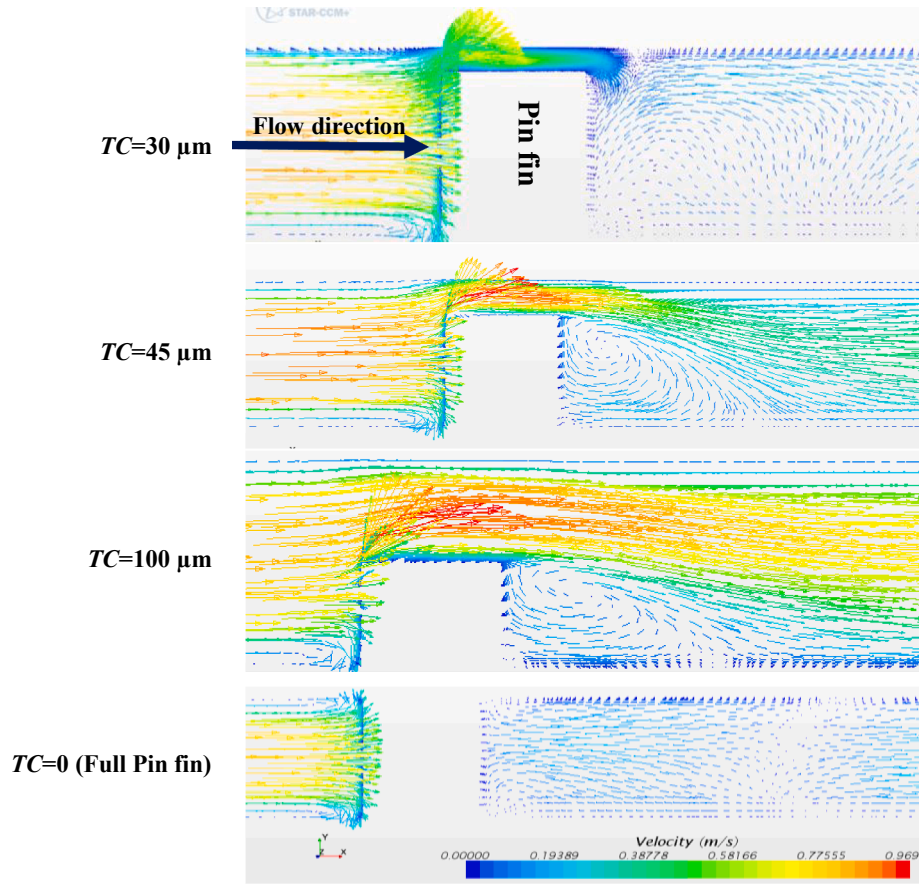


Fig. 11. Velocity vectors colored by velocity magnitudes at the middle longitudinal cross section for $Re = 395$.

sizes from 100, 90 and 80 μm , which resulted in a total grid element count of 550,298, 1,380,684 and 2,897,726, respectively. A maximum of 4% difference equivalent to about 1 $^{\circ}\text{C}$ was observed for the heater average temperature between the medium resolution grid and the fine grid. Therefore, to save computational time, medium grid was used for the simulations.

3. Results and discussion

The experiments and numerical simulations provided insight about heat transfer and fluid flow processes as a function of TC and Reynolds number.

3.1. Effect of tip clearance

Fig. 8a shows the temperature difference for $Re = 395$ and relative TC of $TC/H = 0, 0.15, 0.225$ and 0.5 corresponding to TC of 0, 30, 45 and 100 μm . A TC of $TC/H = 0.15$ experienced a higher temperature

differences than the full height pin fin, and for $TC/H = 0.225$ the surface temperatures were about the same as the full height pin fin. However, for $TC/H = 0.5$, the temperatures were much lower than for the full height pin fin. At certain locations (e.g., $x = 450, 600 \mu\text{m}$, equivalent to $x/D = 3$ and 4, respectively) a lower surface temperature differences of around 10 $^{\circ}\text{C}$ was observed for $TC/H = 0.5$. Close to the pin fin, the temperature difference between the 100 μm TC and the full-length pin fin increased downstream. It then reached a maximum around $x/D = 3-4$ before starting to taper off until about $x/D = 10$.

Fig. 8b depicts the associated heat transfer coefficient of the abovementioned experiments. As expected and following the surface temperature trends, the TC of 100 μm had the highest heat transfer coefficient. A maximum heat transfer coefficient of 30,667.5 $\text{W}/\text{m}^2\text{K}$ was observed with a TC of 100 μm at $x = 450 \mu\text{m}$, compared to a heat transfer coefficient of 18,633 $\text{W}/\text{m}^2\text{K}$ for the pin fin without TC. At $x = 1500$ ($x/D = 10$) μm , the heat transfer coefficient of all pin fins except for the TC of 100 μm converged to an almost single heat transfer coefficient value of about 17,800 $\text{W}/\text{m}^2\text{K}$ while the heat transfer

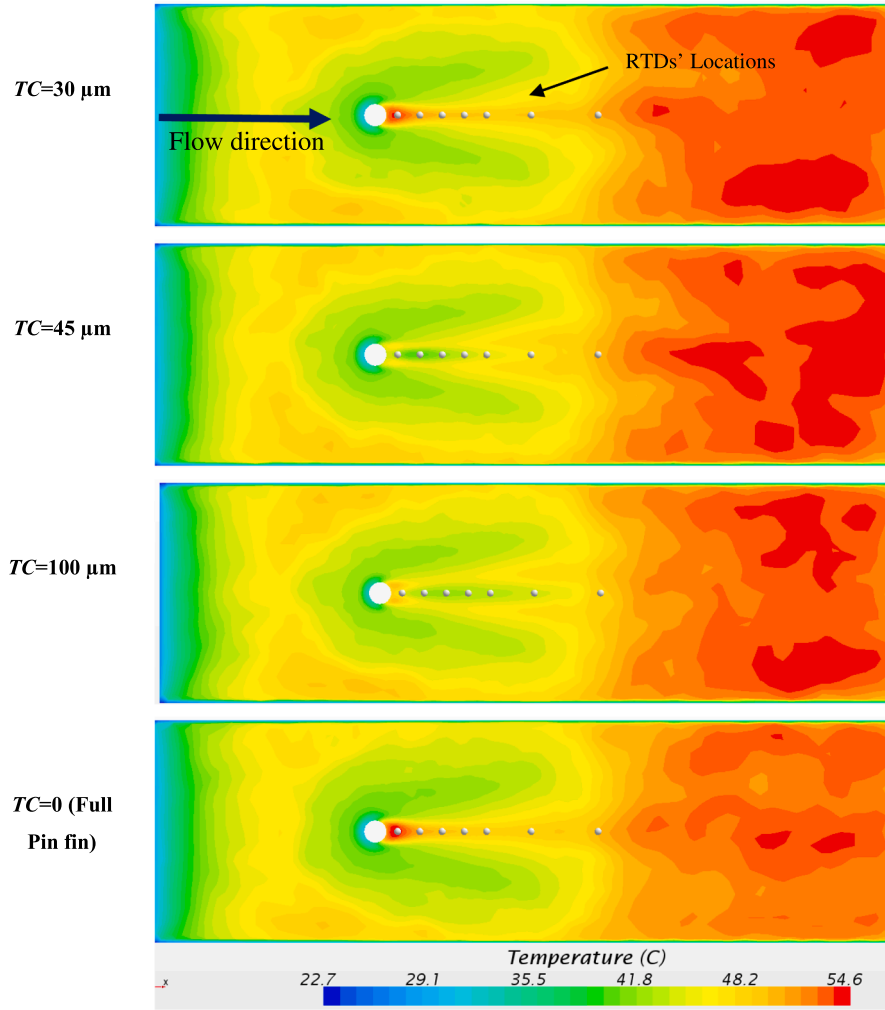


Fig. 12. Contour of heater surface temperature ($Re = 395$).

coefficient of $TC = 100 \mu m$ was still higher and equal to $21,637 \text{ W/m}^2\text{K}$. Fig. 9c depicts the Nusselt number as a function of x/D . Similar to the heat transfer coefficient, the local maximum Nusselt number of $Nu_x = 18$ was obtained for $TC/H = 0.5$ at $x/D = 3$.

Fig. 9 compares the temperature differences obtained by numerical modeling against experimentally measured data on RTD locations and shows that the simulation results are either inside or very close to the measurements uncertainty ranges.

The numerically simulated top view-velocity magnitudes and side-view velocity vectors are shown in Figs. 10 and 11, respectively. As the TC increased from zero to $100 \mu m$, the wake region weakens (higher velocities) and contracted causing the recirculation zone to diminish. This can be related to the flow structures that developed by a

sufficiently large gap on top of the pin fin. Induced accelerated flow on the pin fin top and its interaction with the wake behind the pin fin formed a three dimensional flow, often termed downwash flow, that resulted in enhanced mixing and reduced wake region downstream the pin fin. The tip gap flow was pulled into the pin fin's wake, and thereby disrupted the formation of the recirculating zone. This subtle and hard to identify effect can be seen in Fig. 11 illustrating the velocity vectors in the middle longitudinal cross section of the channel. Surface temperature, heat flux, and heat transfer coefficient contours at a distance of $50 \mu m$ from the heater are shown in Figs. 12–14. The modified wake region associated with a TC of $100 \mu m$, shown in Figs. 13–14, was responsible for the lower surface temperatures and higher local heat flux.

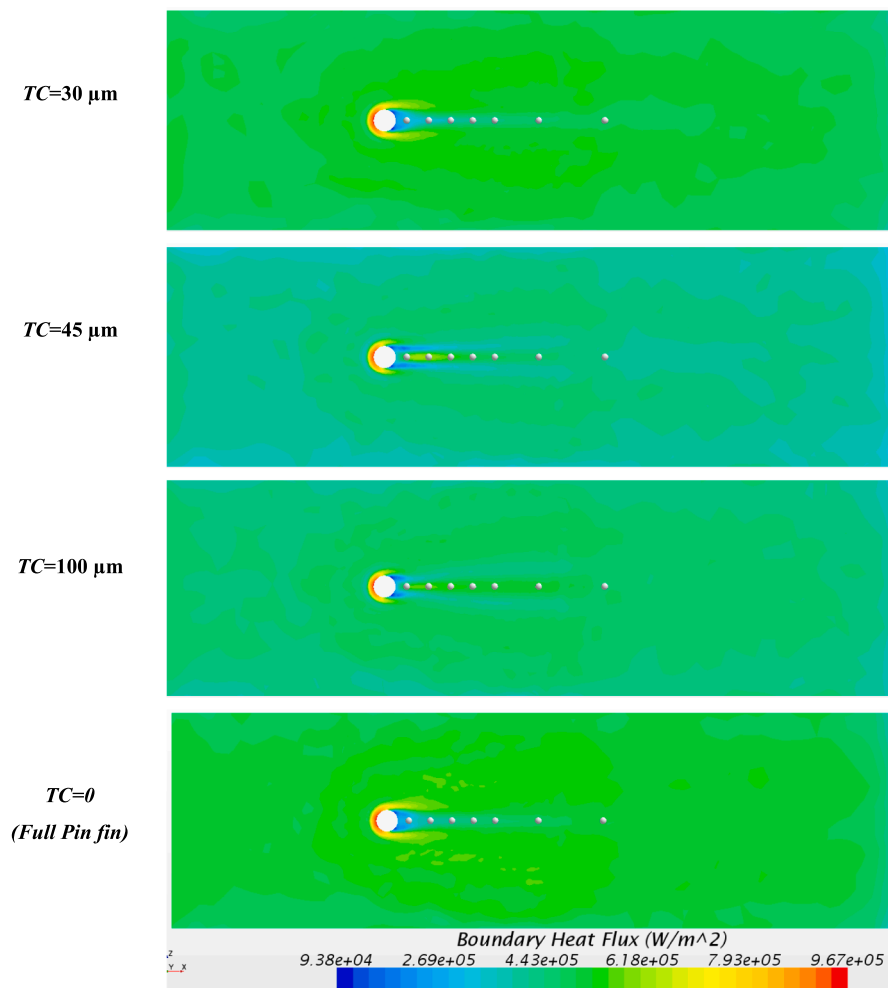


Fig. 13. Contour of heater surface heat flux ($Re = 395$).

3.2. Reynolds number effect

The effects of Reynolds number were studied for EXP5-EXP8 and EXP9-EXP12 (see Table 1), corresponding to Reynolds numbers of 627 and 827, respectively. The same TC s of 0, 30, 45 and 100 μm were examined. Temperature measurements from these experiments and their corresponding calculated heat transfer coefficients and Nusselt numbers are shown in Figs. 15 through 17. Similar to the lower Reynolds number of 395, a $TC = 100 \mu\text{m}$ showed lower temperatures and higher heat transfer coefficients for both Reynolds numbers. With increasing Reynolds number, the Nusselt number increased. Similar to the lower Reynolds number of 395, a TC of 100 μm showed lower temperatures and higher heat transfer coefficients/Nusselt numbers for

both Reynolds numbers of 627 and 827. Temperature difference for full pin fin showed a maximum variation of about 3 $^{\circ}\text{C}$ for both $Re = 627$ and $Re = 827$ (the same trend was obtained for $Re = 395$, as shown in Fig. 8). This could be due to the absence of downwash effect in the near-wake region of the full pin fin as depicted earlier in Fig. 11. Temperature difference trends for TC cases remained almost the same for both Reynolds numbers. In the near-wake area (i.e., $x = 150$ and $300 \mu\text{m}$), the temperature difference for $TC = 45 \mu\text{m}$ was lower than $TC = 100 \mu\text{m}$, which might be due to higher accelerated velocity on the pin fin top for $TC = 45 \mu\text{m}$ compared to $TC = 100 \mu\text{m}$. For the far-wake, overall temperature differences were higher for $TC = 100 \mu\text{m}$.

A detailed view of the wake region is depicted in Fig. 18 where a line integration convolution method in the numerical model was used

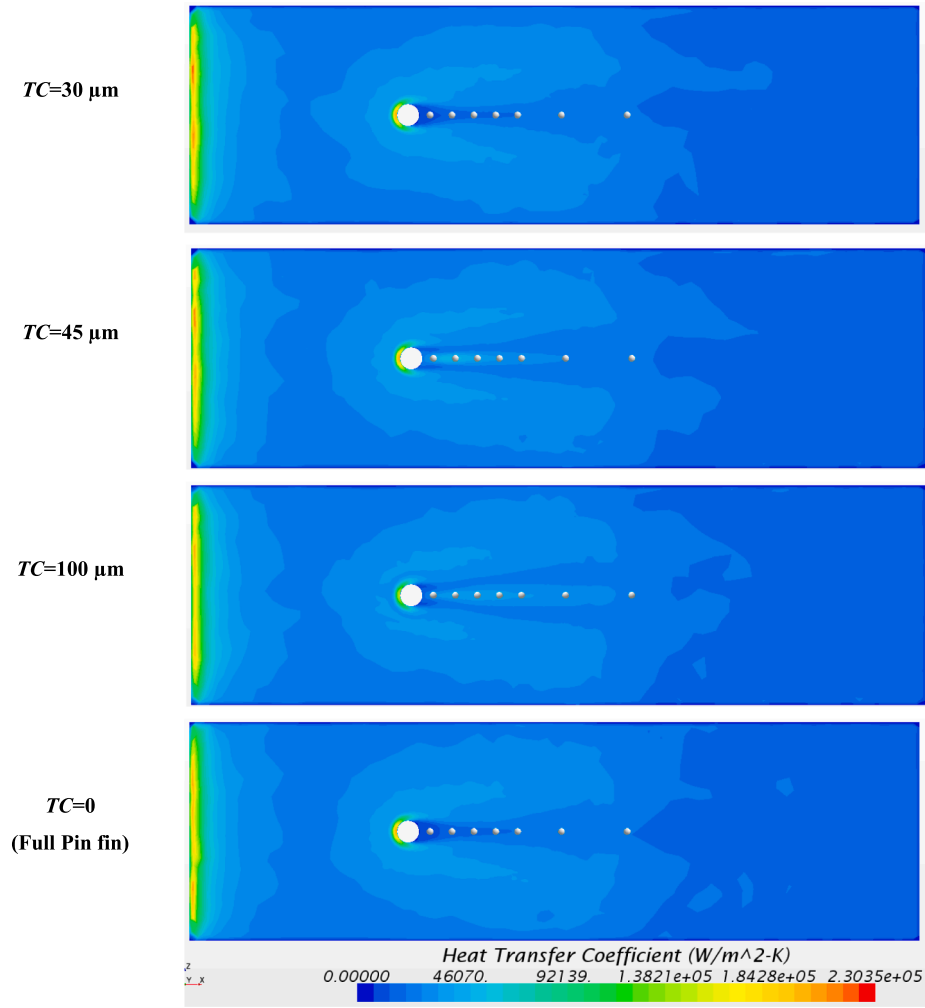


Fig. 14. Contour of heater surface heat transfer coefficient ($Re = 395$).

to obtain a more vivid velocity vectors and streamlines. The wake region consisted of two recirculation zones that were initiated from the pin fin and from a closed low-velocity region with a downstream trailing edge. In Fig. 19, dimensionless wake lengths are plotted against the Reynolds number for various TCs and full pin fin configurations. Except for $TC = 30 \mu\text{m}$ ($TC/H = 0.15$), the wake lengths for pin fins with a TC were shorter compared to the full pin fin. This is in agreement with previous study by Rozati et al. [29]. The shortest wake lengths for all the Reynolds numbers corresponds to $TC = 100 \mu\text{m}$ ($TC/H = 0.5$), which as discussed earlier, is associated with the highest Nusselt numbers.

4. Conclusion

Heat transfer and fluid flow inside a $200 \mu\text{m}$ tall microchannel with a pin fin and a TC was experimentally and numerically studied. Multiple set of devices with TCs of 0, 30, 45 and $100 \mu\text{m}$ were made from SU8 and were sandwiched between two Pyrex wafers. The devices were capable of measuring local temperatures through embedded RTDs on top of the heater surface downstream and around the pin fins. A series of twelve experiments using four devices with different TCs for three different Reynolds numbers of 395, 627 and 827 based on the hydraulic diameter of the channel were conducted. The local temperatures and their corresponding heat transfer coefficient/Nusselt numbers along the centerline of the pin fin were presented and

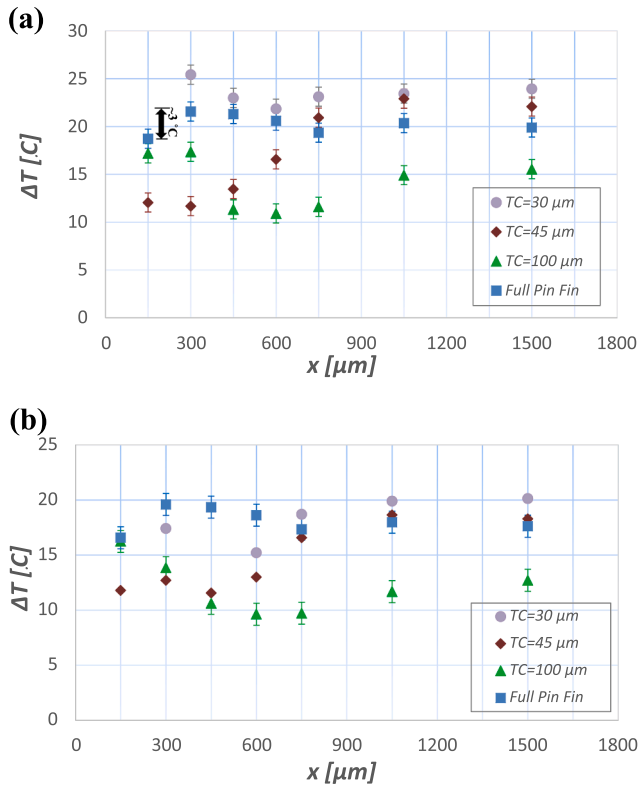


Fig. 15. Effect of Reynolds number on local surface temperature differences for various TCs for $q'' = 53 \text{ W/cm}^2$, (a) $Re = 627$, (b) $Re = 827$.

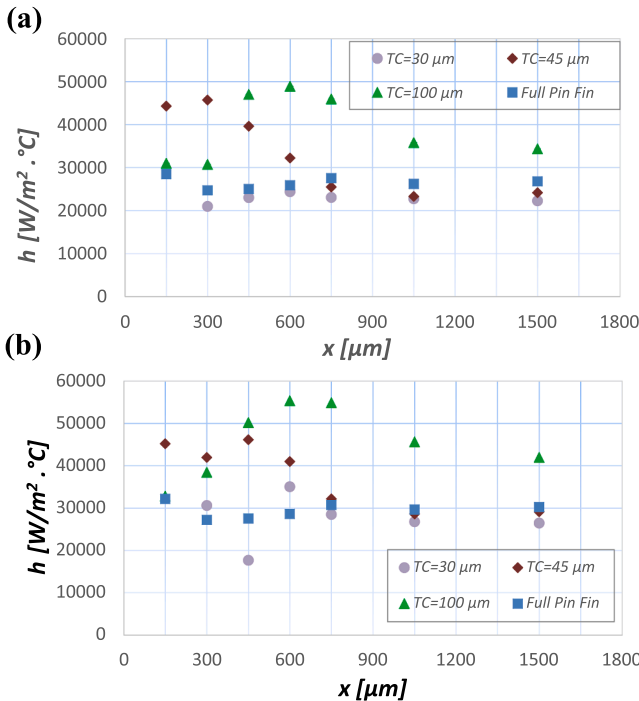


Fig. 16. Effect of Reynolds number on the local heat transfer coefficient in different locations downstream the pin fin.

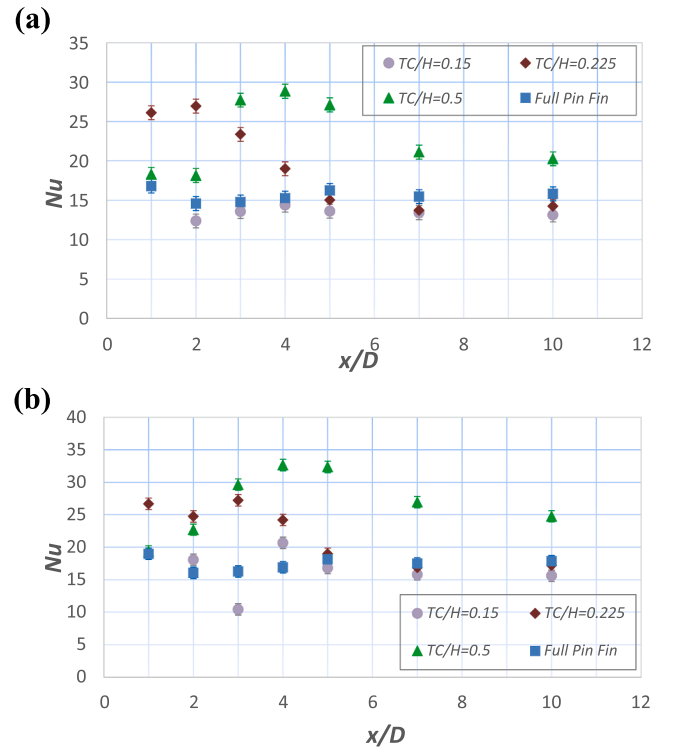


Fig. 17. Effect of Reynolds number on local Nusselt number for various TCs for (a) $Re = 627$ and (b) $Re = 827$.

discussed. Velocity contours of the flow in the horizontal and the vertical sections of the microchannel along with surface temperature and heat transfer coefficient over the heater surface were shown and discussed. Furthermore, Nusselt number as a function of dimensionless distance for various TCs and the wake length as a function of Reynolds numbers were examined. Results were compared with the full height pin fin for reference.

Both experimental and numerical results showed a significant enhancement in heat transfer for a sufficiently large TC in the wake region. The numerical simulations revealed that the TC altered the flow structure downstream the pin fin, augmented the three dimensionality and mixing of the flow, shortened the wake region and increased the velocities in that area. In addition, results suggest that among all the various TCs of the current study, a relative TC of 0.5 enhanced heat transfer the most. Relative TCs of 0.15 and 0.225 did not show a noticeable improvement in heat transfer.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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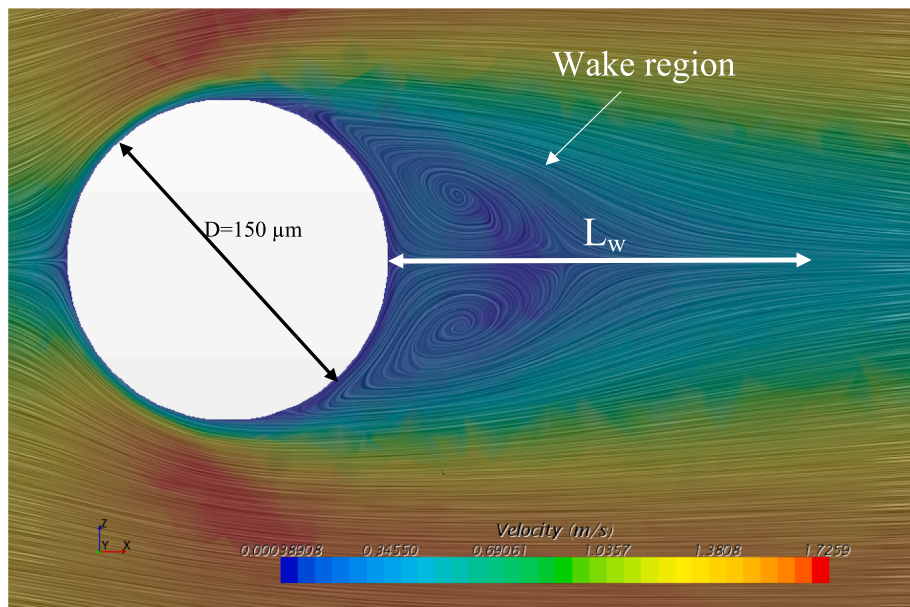


Fig. 18. Wake region for $TC = 45 \mu\text{m}$ at $Re = 395$.

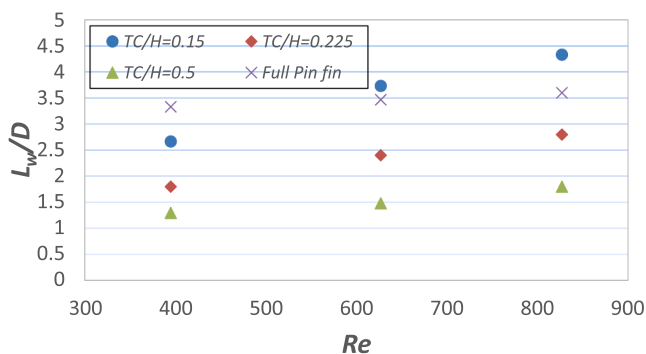


Fig. 19. Dimensionless wake length as a function of Reynolds number—wake length were taken at a distance of $25 \mu\text{m}$ above the heater surface.

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